Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=4;i++)
4       (i-1)[arr] = (i%4);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 2;
8   }
9  int main(){
10   int arr[5] = {1,1,1,1,1};
11   func1(arr+1);
12   for(int i=0; i<5; i++)
13          printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<5; i++)
17          printf("%d ", i[arr] );
18   return 0;
19 }
```

long & long int + int

a[0] = 1
a[1] = 1
a[2] = 2
a[3] = 3
a[4] = 0

arr[1] = a[0] = 1

Output
1 1 2 3 0
2 1 2 3 0
Will the answer change if we write func2 as: **void func2 (int a[1] , int arr){** **int temp = arr + 2;** **a = &temp;** **}** Explain your answer

Q. 2: Write the output of the following sequence of instructions **long arr2[60];** (int) (8)
int arr1[50], a1=(&arr2[40]), a2=(&arr2[30]), b1=(&arr1[40]), b2=(&arr1[30]);
printf("%d %d", a1-a2 , b1-b2); Explain your answer. (2+2 Marks)

Output: 80 40
in first case arr2 is long int and sizeof(long int is 8 bytes. so $a_1-a_2=8\times10$
in second case arr1 is int type and sizeof int is 4 bytes. $b_1-b_2=4\times10$.